Challenge 1: finding the real cause

#!/usr/bin/env python3

The substring starts at offset 6. The old rule checked offset 1.

OLD RULE: WRONG POSITION

```
/bin/env python
at offset 1
```

application/x-sh

FIXED RULE: START OF FILE

```
#!/usr/bin/env python
at offset 0
```

text/x-python

One prefix covers python, python3 and python3.11 without a space after #!.

Challenge 2: proving the fix

01 Separate content from filename hints

Test each input without a filename, then with script.py.
This checks the content rule and the effect of metadata.

02 Keep working behavior intact

Check direct Python paths, Shell scripts and an input
with a newline before the shebang.

03 Same 16 focused checks: 6 failures before, 0 after

All 16 passed after the fix. These are local results.
They do not establish remote CI or whole-project validation.

Negative tests catch false positives

x/bin/env python

x is at offset 0. /bin/env python begins at offset 1.

OLD RULE: ACCIDENTAL MATCH

Matches the text after x
Mistakes ordinary text for Python

text/x-python

EARLIER FIX: FILE-START CHECK

No #! prefix at the start
The Python rule does not match

text/plain

Kyubin checked the earlier fix separately. A permanent regression test is deferred.

What I learned and would change

Understand rule interactions	Trace competing rules and write the expected behavior before changing the implementation.
Plan broader coverage earlier	Build an initial test matrix for Python, Shell, filename hints and boundary cases.
Preserve useful review checks	Turn probes such as x/bin/env python into permanent automated regression tests.
Keep the useful workflow	Continue reproducing first and comparing the same checks before and after a focused fix.